

Skills Summary

Building and Comparing Two Number Patterns

Use this reference while completing your worksheet or when studying.

A rule says where to start and what to add. Two rules make two patterns, and the interesting question is how the *matching* terms of the two patterns are related.

Building a pattern from its rule

Rule: Write the starting number as term 1. Add the step to get each new term.

Going from term 1 to term n takes $n - 1$ steps, so term $n = \text{start} + \text{step} \times (n - 1)$. Count the steps, not the terms: five terms means four additions.

Example: Start at 1 and add 4 each time. What is term 5?

Step 1. Term 5 is four steps after term 1, so add the step four times rather than five — the start is already term 1.

Step 2. $1 + 4 \times 4 = 1 + 16$. Written out: 1, 5, 9, 13, 17.

Term 5 = 17

Try it: Start at 2 and add 5 each time. What is term 6?

check: 27

Comparing corresponding terms

Rule: Corresponding terms are the terms in the same place. Line the two patterns up in a table, one under the other, then divide a term of the second by the matching term of the first.

When both patterns start at 0, the answer is the same for every pair, and it equals the second step divided by the first. When the patterns start somewhere else, doubling the step does *not* double the terms unless the starting number doubles too.

Example: Pattern A: 0, 3, 6, 9. Pattern B: 0, 9, 18, 27. How many times B's term is A's term?

Step 1. Both patterns start at 0, so pick any matching pair and divide — B on top, because the question asks what A was multiplied by.

Step 2. Fourth terms: $27 \div 9$. Checking another pair, $18 \div 6$ gives the same.

Each B term is 3 times its A term

Try it: Pattern A: 0, 5, 10, 15. Pattern B: 0, 20, 40, 60. How many times B's term is A's term?

check: 4

Turning two patterns into ordered pairs

Rule: Take corresponding terms and write them as (first pattern's term, second pattern's term). Plot each pair by going across first, then up.

The points always fall in a straight line, and the relationship between the matching terms tells you the second number from the first one.

Example: Pattern A: 0, 2, 4, 6. Pattern B: 0, 10, 20, 30. Pairs: (0, 0), (2, 10), (4, 20), (6, 30). What is the second number of the pair whose first number is 6?

Step 1. Each B term is $10 \div 2 = 5$ times its A term, so the second number of any pair is 5 times the first — that is quicker than extending the table.

Step 2. $5 \times 6 = 30$, which matches the fourth pair already listed.

Second number = **30**

Try it: Pattern A: 0, 3, 6. Pattern B: 0, 12, 24. What is the second number of the pair whose first number is 9?

98 :xpəp

(!) Watch out: Term 1 is the starting number, before any adding — so term 6 takes five additions, not six. And when you plot an ordered pair, the first number is the across number every time.